

Microtubular Screw Symmetry: Packing of Spheres as a Latent Bioinformation Code

DJURO L. KORUGA

Molecular Machines Research Unit

University of Belgrade

11000 Belgrade, Yugoslavia

and

University of Arizona Health Sciences Center

Tucson, Arizona 85724

INTRODUCTION

Microtubules (MT) participate in bioinformation processes such as memory and learning.¹⁻² The precise role of MT in these functions is not well understood. If MT are primary information processors, then their subunits should be arranged by sphere-packing symmetry derived from information coding laws.

THE PACKING OF SPHERES

Oh ($\bar{6}/4$) symmetry group describes face-centered-cubic sphere packing and derives information coding laws.³ Hexagonal packing of protein monomers independent of the Oh ($\bar{6}/4$) symmetry group has been used to explain the form patterns of viruses, flagella, and MT.⁴ Because hexagonal packing and face-centered-cubic packing of spheres have equal density, I use both to explain MT organization.

Hexagonal packing may be used by fixing conditions. This is possible if the centers of spheres lie on the surface of a cylinder (with radius equal to the Oh ($\bar{6}/4$) unit sphere) and if (and only if) the sphere values in the axial direction (lattice) of the cylinder by order of sphere packing is the same as in the dimension in which the face-centered-cubic packing is done.

Because $\bar{6}$ -fold symmetry axis of Oh group is inverse, there must be two kinds of spheres (white and black) on the cylinder surface, but linked such that they have the dimension value in which the face-centered-cubic packing is done. This packing on the cylinder surface leads to "screw symmetry" (FIGURE 1).

From coding theory, the symmetry laws of α - and β -tubulin subunits lead to the conclusion that 13 protofilaments passes one of the best known binary error-correcting codes [K_1 (13, 2^6 , 5)] with 64 code words.

Symmetry theory further suggests that on the surface of a circular cylinder in axial direction there must be a code of length of 24 monomer subunits. If the coding efficiency is used as a criterion of transmission, then 6-binary dimers of K_1 code should be coded to give a 4-dimer ternary sequence of K_2 [24, 3^4 , 13] code. This code may result from interaction between 24 tubulin monomers and high molecular weight proteins. Under the influence of Ca^{2+} -calmodulin, binary dimers of K_1 code give dimer ternary sequence of K_2 code. In this way, K_1 and K_2 codes, which result from the property of the Oh ($\bar{6}/4$) symmetry group, lead a K (B^6T^4) transmission bioinforma-

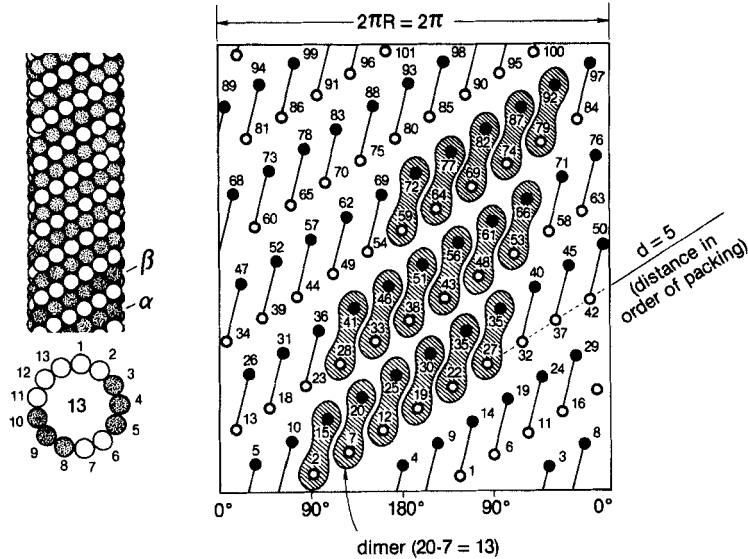


FIGURE 1. Because the optimal number of spheres for information processing is 11, 12, or 13, we show here the solution for the packing of 13 spheres. The distance between spheres in order of packing is 5. Sphere indices in axial direction of dimers (white-black) must be 13 because each dimer represents the dimension of face-centered-cubic packing. From information theory, this means that the arrangement of white and black spheres corresponds to a binary code of length $n = 13$ and distance $d = 5$ with 64 code words.

tion code, which may combine with guanosime triphosphate (GTP) and ions for intracellular dynamic and organizational activities.

CONCLUSION

Symmetry theory and MT structures lead to the conclusion that packing of tubulin subunits is equal to information coding. This means that microtubules possess a code

TABLE 1. K-function Code

Digital Sum	Code	
	$K_1[13, 2^6, 5]$	$K_2[24, 3^4, 13]$
0	18	19
1	32	$16^+ \quad 16^-$
2	10	$10^+ \quad 10^-$
3	4	$4^+ \quad 4^-$
4	—	$1^+ \quad 1^-$
All	$2^6 = 64$	$3^4 = 81$

Coding relations between K_1 and K_2 codes show that 18 binary words of K_1 code yield 19 ternary words of K_2 code; there exists one “nonsense” word. Further, we see that 14 binary words are ambiguous with ternary words. Finally two more “nonsense” (stop) and ambiguities are seen in coding of amino acids in the genetic code. Thus, MT may code for dynamic information processing in some elements like DNA codes for genetic structure.

system that can provide both memory and intracellular dynamic activities. The latent bioinformation K code has been identified and in that way showed that microtubules along with DNA and RNA are unique cell structures that possess a code system.

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